The answer keys, explained: ATC, FDA's classes through RxNav, OnSIDES and the pre-registered pairs

Written 2026-10-15 against the active build of 2026-10-13 (2,452 entries). **An explainer: it adopts nothing and changes nothing.** It says what each outside source is, how this project turns it into an answer key, and what each key can and cannot tell you. The measurements, and the arguments for each key, are in SIMILARITY-HARNESS.md §3–4 and §13–16 and in SIMILARITY-LITERATURE.md Parts 3–4. Where a figure here disagrees with CLAUDE.md's **Status**, Status is right.

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0. What an answer key is for

The page ranks every molecule against every other by shared targets and shows the closest. Nobody can check that ranking pair by pair: there are 2,452 entries and about three million pairs. So the ranking is graded against **answer keys**: lists, written by someone else for another purpose, that say which drugs belong together.

Every key here is used the same way. For each molecule the key can ask about (a **question**), the harness asks the page for its similarity list and walks down it to the first molecule the key calls a **classmate**. How far down that is is the molecule's **rank**. The grade summarises the ranks across all the questions (§6 defines each figure).

A key is worth more the less it shares with the score. If the score reads ChEMBL's curated mechanisms and the key restates the same mechanisms, a good grade partly measures the score agreeing with itself. That is why there are several keys from different curators, and why the table in §5 says what each one shares with the score.

key	what it says belongs together	curator	section
ATC level 4 (the headline) and its repair	drugs WHO files in one subgroup	WHO	§1
FDA EPC and PE	drugs FDA gives one pharmacologic class or physiologic effect	FDA, reached through NLM's RxNav	§2–3
label effects	drugs whose labels list similar nervous-system side effects	drug labels, read by OnSIDES	§4
pre-registered pairs	243 named drug pairs: alike, unlike, or opposite	written here, each with a cited source	§5

RxNav is not a key. It is the identifier service the FDA and OnSIDES keys both go through, so it gets its own section (§2).

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1. ATC: the WHO's drug classification, and its levels

What it is

The **Anatomical Therapeutic Chemical (ATC) classification** is the WHO's filing system for drug substances. The WHO Collaborating Centre for Drug Statistics Methodology in Oslo maintains it and revises it every year. It was built for counting drug use across countries, not for describing pharmacology, and that purpose shows in how it groups drugs.

Each code is a nested address with five levels. Amitriptyline is `N06AA09`:

level	code	name	what WHO groups by at this level
1	`N`	Nervous system	the anatomical main group: the organ or system the drug acts on (14 groups, A to V)
2	`N06`	Psychoanaleptics	a therapeutic or pharmacological subgroup
3	`N06A`	Antidepressants	a chemical, pharmacological or therapeutic subgroup
4	`N06AA`	Non-selective monoamine reuptake inhibitors	a chemical, pharmacological or therapeutic subgroup
5	`N06AA09`	amitriptyline	the chemical substance

Three properties matter for using it as a key.

- **A drug can have several codes**, one for each use WHO recognises. Caffeine is `N06BC01` (a psychostimulant, *xanthine derivatives*) and `D11AX26` (a dermatological). Morphine carries `N02AA01` and two codes for combination products.
- **Levels 3 and 4 group by whatever WHO found most useful for that subgroup**: chemical structure, mechanism or use. WHO's own definition says "chemical, pharmacological or therapeutic". The same benzodiazepines are filed twice, as anxiolytics (`N05BA`) and as hypnotics (`N05CD`). Every opioid agonist sits in one of five structure classes (`N02AA` to `N02AF`), and every D2-blocking antipsychotic in one of nine. Of the 64 level-4 classes the key uses, 42 group by chemical structure, 16 by mechanism and 4 by use, read by class name (SIMILARITY-LITERATURE §3.5).
- **A fourth-level letter `X` means "other"**: `N06AX` is *Other antidepressants*, a group defined by its members **not** sharing a mechanism (mirtazapine and bupropion are both in it).

How it becomes the headline key (`--key atc`)

1. ChEMBL's `molecule_atc_classification` is read at build time, keeping **only N codes**. That filter is the cohort's ATC admission route; it was never designed to define a key. The codes are stored in `molecules.atc_level5` and exported in `viz/graph.data.js`.
2. The harness truncates each code to five characters, its level-4 class (`N06AA09` → `N06AA`), drops every `X` class, and drops classes with only one member.
3. Two molecules are classmates when they share a level-4 class. That gives **395 questions in 64 classes**.

On the current build, 284 of the 395 have a classmate somewhere in their list and are graded: **top-1 32.7%, MRR 0.4607**. The other 111 have no classmate anywhere in their list, which says something about ChEMBL's coverage rather than about the ranking, so they are left out of the rates.

What it gets wrong. Because it reads only N codes, a right answer filed elsewhere counts as a miss. Naloxone is `V03AB` (*antidotes*), so it is wrong as naltrexone's nearest neighbour; seltorexant has no ATC code, so it is wrong as suvorexant's. Because it is binary, thioridazine → chlorpromazine (two phenothiazine subgroups, `N05AC` and `N05AA`) scores the same as an unrelated drug. And it has no negatives: nothing in it says two drugs must *not* be near each other.

The repaired key (`--key atc-repaired`, v1.0.0)

Built 2026-10-13 and reported beside the headline, never in place of it. Its rules are in `analysis/similarity_harness/atc_repaired.json`:

- the questions are still the molecules with an N code, but **classmates can come from any ATC code** the catalogue holds (read from `molecule_classes`, source `atc`, in the database);
- residual classes filed under a letter other than X (`N02BG`, `N05CM`), classes defined by use (drugs for nicotine, alcohol or opioid dependence, antivertigo preparations, antidotes) and non-pharmacological classes are set aside, each with its reason written next to it;
- a molecule in a **sibling class** (same level 3) earns half credit in nDCG@10. Top-1 and MRR still count only exact classmates.

It asks 353 questions, grades 259, and scores top-1 36.3%, MRR 0.4958 and nDCG@10 0.3850.

What it shares with the score: partly the curated mechanisms. Where an ATC class follows mechanism (SSRIs, MAO inhibitors), it encodes the same knowledge ChEMBL's curators write into `drug_mechanism`, and 39 of the 110 top-1 hits on the 1,335-entry build were won mainly at a target where both molecules have a curated mechanism (SIMILARITY-LITERATURE §3.4).

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2. RxNav, RxNorm and RxClass: how a molecule reaches FDA's labels

What they are

RxNav is the U.S. National Library of Medicine's browser and free public web API for its drug terminologies. Two of them are used here:

- **RxNorm** gives every clinical drug sold in the US a normalised name and an identifier, the **RxCUI**. Its concepts come in term types. The ones that matter here:
  - **IN**, an *ingredient* (sertraline);
  - **PIN**, a *precise ingredient*, a salt or ester (sertraline hydrochloride);
  - brand names and clinical drugs (a strength and dose form), which this pipeline never uses.
- **RxClass** links RxNorm ingredients to drug classes from several class systems: ATC, MeSH, the VA's classes and, the ones used here, FDA's pharmacologic classes (§3).

A molecule in this catalogue is a ChEMBL structure, not an RxNorm drug, so something has to connect the two. That something is the **FDA UNII** (Unique Ingredient Identifier). FDA's Global Substance Registration System assigns it to a substance from its structure, and both FDA and RxNorm index by it.

The route, an identifier at every step

`scripts/resolve_cohort_fda_classes.py` goes from structure to class without ever searching by name (invariant 15):

```
molecules.inchikey ──► FDA UNII ──► RxCUI ──► RxNorm ingredient (IN) ──► FDA classes
  (the structure)      (FDA's UNII     (RxNav,     (a PIN is followed to      (RxClass, asserted
                       records file,    by UNII)    its one IN; any other      for the ingredient
                       exact InChIKey)              term type is unused)       itself)
```

Three rules keep it from guessing:

- **Exactly one ingredient, or nothing.** A molecule whose identifiers reach two ingredients is dropped as ambiguous and listed, never resolved to either. A wrong drug is worse than a missing one (invariant 6).
- **The parent's own structure decides.** ChEMBL folds salts and formulations onto a parent; RxNorm sometimes keeps such a form as a drug of its own. ChEMBL folds aminophylline onto theophylline, but RxNorm files aminophylline as a separate ingredient (RxCUI 689 against 10438). Reading every form at once dropped theophylline as ambiguous, so a folded form is now read only where the parent reaches nothing (DATA-QUALITY-LOG, 2026-10-13).
- **Not pinned, but cached.** RxNav is a live service with no downloadable release this pipeline can pin. Every answer is cached as JSON under `db/cache/rxnav/`, a rerun is offline, and RxNav's version and the run date are recorded in `build_manifest`. A cold run takes about 50 minutes.

Coverage. 930 molecules reach exactly one ingredient: 911 through the parent and 19 through a form. 909 molecules have a UNII that RxNorm does not know, most presumably because they were never sold in the US. Nobody has checked.

The OnSIDES key (§4) reuses this route from the same cache, because OnSIDES is keyed by the same RxNorm ingredient ids.

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3. FDA's pharmacologic classes: EPC, MoA and PE

What they are

When FDA approves a drug label, it indexes the active ingredient with pharmacologic classes on four axes, three of which are read here. The terms come from **MED-RT** (Medication Reference Terminology), the Department of Veterans Affairs' drug-class terminology that replaced NDF-RT and that FDA draws its class terms from. That is why every class code here looks like `N0000175696`.

- **Established Pharmacologic Class (EPC).** The class FDA judges scientifically valid and clinically meaningful for the drug's approved indication. It is the phrase in the indications line of a label's *Highlights*: "Zoloft is a *selective serotonin reuptake inhibitor (SSRI)* indicated for ...".
- **Mechanism of action (MoA).** What the drug does at the molecular level. It mixes pharmacology with pharmacokinetics: sertraline is a *Serotonin Uptake Inhibitor* and also a *Cytochrome P450 2D6 Inhibitor*.
- **Physiologic effect (PE).** What happens in the body. This is the nearest thing among the class keys to an effect axis, and it is coarse.

- The fourth axis, *chemical structure*, is not read.

From this database (the labels are as RxClass serves them):

molecule	EPC	MoA	PE
sertraline	Serotonin Reuptake Inhibitor	Serotonin Uptake Inhibitors; Cytochrome P450 2D6 Inhibitors	Increased CNS Serotonin Activity
amitriptyline	Tricyclic Antidepressant	Norepinephrine Uptake Inhibitors; Serotonin Uptake Inhibitors	Increased CNS Norepinephrine Activity; Decreased Serotonin Degradation
morphine	Opioid Agonist	Full Opioid Agonists; Opioid mu-Receptor Agonists	Analgesia; Decreased GI Motility; and four more
caffeine	Central Nervous System Stimulant; Methylxanthine	Enzyme Inhibitors	Central Nervous System Stimulation; Bronchodilation; and three more

The table shows the axes' character. An EPC is narrow and mostly names a mechanism. A MoA can name one precisely (*Opioid mu-Receptor Agonists*) or say nothing (caffeine's *Enzyme Inhibitors*). A PE can be generic: *Central Nervous System Stimulation* holds both caffeine and naltrexone.

RxClass serves each class with the source that asserted it. Only rows from **FDASPL** (FDA's structured product labels) and **MEDRT** are kept, and only those asserted for the ingredient itself; DAILYMED's rows repeat FDASPL's and are ignored. The backfill stores everything else it reads too, CYP450 classes included, as the `fda_epc`, `fda_moa` and `fda_pe` sources of `molecule_classes`. **The page never shows them.** Their only use is grading.

Coverage. 627 molecules have an EPC, 787 a MoA and 733 a PE. Only drugs with a US label are covered.

How they become keys (`analysis/similarity_harness/fda_keys.json`, v1.0.0)

- `--key epc`. Molecules sharing an EPC are classmates. Molecules sharing only a MoA earn half credit in nDCG, except through MoA classes that describe how a drug is handled (CYP450 enzymes, P-glycoprotein and other transporters, glucuronidation) or that say nothing (*Enzyme Inhibitors*, *Receptor Interactions*, *Unknown Cellular or Molecular Interaction*). Each exclusion is listed with its reason.
- `--key pe`. Molecules sharing any physiologic effect are classmates, unfiltered.

key	questions	graded	top-1	MRR	filtered top-1	filtered MRR
EPC	487	455	45.7%	0.5946	64.0%	0.7319
PE	701	598	37.6%	0.5154	53.8%	0.6498

What they share with the score. FDA is a different curator from WHO and ChEMBL, but an EPC such as *Serotonin Reuptake Inhibitor* states the same mechanism ChEMBL curates. EPC is the narrowest key here and agrees with ATC on little more than half of each other's classmate pairs (SIMILARITY-HARNESS §13).

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4. OnSIDES: side effects from drug labels

What it is

OnSIDES is a database of the adverse effects listed in drug labels, made by the Tatonetti lab (Tanaka et al. 2025, *Med*; GitHub). A fine-tuned language model reads label text from the US (DailyMed), the UK, the EU and Japan and turns each adverse effect it finds into a **MedDRA** term. The result is keyed by RxNorm ingredient. Release v3.1.1 is pinned by sha256 and licensed CC BY 4.0.

It matters here because **it is the only key that shares no input with the score.** Every other key reads targets or a curator's word about a mechanism. A label's adverse-reactions section records what happened to patients. If two drugs' labels list the same unusual nervous-system effects, they may act alike in the brain whatever ChEMBL has measured.

MedDRA, and the hierarchy OnSIDES leaves out

MedDRA (Medical Dictionary for Regulatory Activities) is the standard vocabulary for adverse events. It is a five-level tree:

level	example
System Organ Class (SOC)	Nervous system disorders
High Level Group Term (HLGT)	Movement disorders (incl parkinsonism)
High Level Term (HLT)	Tremor (excl congenital)
Preferred Term (PT)	Tremor
Lowest Level Term (LLT)	Shakiness, Tremulousness, Tremor limb: the wordings a label may use

The key needs only the effects under two SOCs: *Nervous system disorders* and *Psychiatric disorders*. MedDRA's hierarchy is licensed, so OnSIDES ships the terms without it. This project places each term through **ADReCS** v3.3 (Adverse Drug Reaction Classification System, Cai et al. 2015), which publishes MedDRA's tree with its codes. ADReCS is pinned and read locally, but it states no open licence. That is why a second reading that needs no hierarchy, `onsides-all`, sits beside the nervous-system key.

Each term is placed by code where it is a PT. Where it is an LLT, it is placed by its exact name among one PT's synonyms. An LLT that names several PTs ("Convulsion") is never resolved to one of them and stays its own effect. A Wikidata route was tried first and reached only 95 of the nervous-system terms the catalogue's labels use, missing *Depression* and *Hallucination* (DATA-QUALITY-LOG, 2026-10-14).

How it becomes a key (`analysis/similarity_harness/onsides_key.json`, v1.0.0)

These rules were fixed before the key graded anything, and committed as such (9d47202):

1. **A label is read only if all its products are one ingredient.** Combination products are skipped, because their effects belong to the partner drug as much as to this one. 691 molecules have at least one single-ingredient label.
2. **An effect belongs to a molecule if two of its labels list it**, or its only label does. A share of labels was the first rule written, but labels of one drug differ a hundredfold in what OnSIDES reads from them: diazepam's 34 US labels yield a median of 4 effects each, its 13

Japanese labels 49. Under that rule, no effect reached half of diazepam's labels.
3. **Only nervous-system and psychiatric effects count**, and a molecule needs at least five of them to be asked or judged. That leaves 409 questions.
4. **Rare effects weigh more.** Each effect is weighted by its *inverse document frequency*, ln(N / df), where N is the number of ingredients with a profile and df the number listing that effect. An effect on nearly every label weighs nearly nothing. That covers class boilerplate, and also OnSIDES's habit of reading the table heading "Nervous system disorders" as the effect "Nervous".
5. **Similarity is the cosine** of two weighted profiles: the overlap measured as an angle, so a long label does not beat a short one by length.
6. **A molecule's five nearest label neighbours are its classmates**; neighbours six to ten earn half credit in nDCG. Two molecules that are one RxNorm ingredient (haloperidol and its decanoate) share every label, so neither is the other's neighbour.

What it says, and where it is noisy

Read before it graded anything (`node analysis/answer_keys/neighbours.mjs`):

- sertraline's classmates are vilazodone, fluoxetine, escitalopram, citalopram and fluvoxamine;
- morphine's are methadone, oxymorphone, meperidine, oxycodone and hydromorphone;
- diazepam's are clonazepam, lorazepam, estazolam, triazolam and *methadone*, which is noise;
- donepezil's include memantine, by shared indication rather than by mechanism;
- ciprofloxacin's are azithromycin and bexarotene, because a non-nervous-system drug's few nervous-system effects are generic.

key	questions	graded	top-1	MRR	top-1 if the list were shuffled
label, nervous system	409	346	14.5%	0.2620	1.8%
label, every effect	609	507	19.3%	0.3180	2.8%

The absolute grade is low, and the key is why. Each question has five classmates out of hundreds, where a class key gives dozens, so a shuffled list almost never hits. Measured as a ratio over the shuffle, the ranking does as well against this key (8.1×) as against the keys that share its curation (ATC 6.8×, EPC 9.0×). The richer a label, the more the ranking agrees with it: filtered top-1 rises from 19% for molecules with 5–8 effects to 37% for those with 17 or more.

Three cautions. Labels carry class boilerplate: every antidepressant label has the suicidality warning. Effects are confounded by indication: drugs for one disease list that disease's symptoms. And the ADReCS licence is unresolved.

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5. The pre-registered pairwise key

What "pre-registered" means

In clinical trials and experimental psychology, a study is **pre-registered** when its hypotheses and its analysis plan are written down and time-stamped before any data are seen. Results cannot then be dredged for whatever looks good afterwards. The same idea applies here: a hand-written answer key could be bent to say anything if it were written after looking at the ranking.

SIMILARITY-HARNESS §3 refused a hand-written key for exactly that reason. `analysis/answer_keys/pairs.json` is hand-written anyway. It answers the objection by how it was made, not by what it says:

- **drafted blind.** Three subagents wrote it, one per area (psychiatry; neurology and pain; autonomic and other receptor systems). Each was given only the catalogue's ChEMBL ids and names and was forbidden to open the repository or run the ranking. The session that commissioned them reviewed the drafts, and every edit it made is listed in the file;
- **sourced.** Every pair cites what it rests on: 207 sources, each a DOI whose title was checked against Crossref or a US label on DailyMed whose setid resolves, quoted with what it says. ChEMBL and GtoPdb (the score's own inputs), DrugBank and Wikipedia were not allowed;
- **frozen.** The pairs and the rules that grade them were committed and pushed (3ed013f) before either was run against the ranking;
- **held out.** No `--key` loads it, so no sweep over the scoring constants can be tuned to it. Changing a pair or a rule makes a new version, logged in SIMILARITY-HARNESS.md, and never in the same change as a constant it judges.

What it holds

243 pairs of three kinds:

kind	count	meaning	example
positive	116	the same primary mechanism at the same target, in the same direction	fluoxetine and sertraline: both inhibit the serotonin transporter
negative (known-unlike)	80	the same indication reached by different primary mechanisms, with no shared primary target	sertraline and bupropion: both for depression; one blocks serotonin reuptake, the other noradrenaline and dopamine uptake and "not serotonin" (its label)
opposite	47	one target, opposite directions	flumazenil and alprazolam: alprazolam enhances GABA-A through the benzodiazepine site; flumazenil competitively blocks that site

Two flags mark pairs to read with care. **`seen`** (44 pairs) marks a relationship the older 50-probe diagnostic had already put to the ranking, so its answer was known in advance. **`doubt`** (34 pairs) marks weaker sourcing or a weaker claim: a class-level source, a mechanism a label calls unknown, or an indication that overlaps rather than matches.

The rules

- Each pair is read **both ways**: b in a's list, and a in b's.
- A **positive** is a hit when the partner ranks within the **top 10**. A partner absent from the list shares no target and cannot be reached; it is counted as coverage, never as a miss.
- A **negative** or an **opposite** is **clean** when the partner ranks below 10 or is absent.
- **Ordering:** in each list asked about at least one positive and at least one negative or opposite, the share of (positive, other) pairs in which the positive ranks higher. An absent partner ranks last, and a tie counts half.
- Three readings are always reported: every pair, without `seen`, and without `doubt`. None replaces another.

First grade, and what it found

`node analysis/answer_keys/pairs.mjs`, on the current build:

reading	positives within top 10	negatives clean	opposites clean	positive above other
every pair	75.4% of 232	98.1% of 160	90.4% of 94	96.0% of 93 lists
without `seen`	77.6% of 156	98.1% of 160	92.7% of 82	95.8% of 72
without `doubt`	77.6% of 210	97.7% of 130	91.0% of 78	96.8% of 82

The `seen` pairs grade no better than the rest, so having been asked before did not flatter the key. Three findings:

- **The opposition rule misses one pair of actions.** 7 of the 9 failed opposite directions are flumazenil against the benzodiazepines and zolpidem, at ranks 2 to 8. ChEMBL curates flumazenil as an ALLOSTERIC ANTAGONIST and them as POSITIVE ALLOSTERIC MODULATOR, and `OPPOSED_ACTIONS` names no such pair. It has not been changed: that list is chosen by argument, and the item is open in CLAUDE.md.
- **Negatives stay apart.** 3 of 160 fail: lacosamide and levetiracetam rank each other 2nd and 6th. Xanomeline ranks olanzapine 10th, and that one reflects a real shared target: olanzapine blocks the muscarinic receptors xanomeline activates.
- **The worst positives** are tranylcypromine to isocarboxazid (rank 183), lacosamide to oxcarbazepine (108), cabergoline to bromocriptine (102) and propranolol to nadolol (101).

What it shares with the score. Labels and reviews state the mechanisms ChEMBL's curators also read, so this key is not independent the way the label key is. What it adds that no other key has is **negatives, opposites and a fixed set of questions**.

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6. The keys side by side

key	judges	shares with the score	questions	graded	top-1	MRR	where
ATC level 4 (headline)	WHO subgroup, N codes only	partly the curated mechanisms	395	284	32.7%	0.4607	`molecules.atc_level5`; `harness.mjs`
ATC repaired	any ATC code, exclusions, sibling half credit	partly the curated mechanisms	353	259	36.3%	0.4958	`atc_repaired.json`
FDA EPC	FDA's established class; MoA for half credit	restates the mechanism	487	455	45.7%	0.5946	`fda_epc`/`fda_moa`; `fda_keys.json`
FDA PE	FDA's physiologic effect	little; a coarse effect axis	701	598	37.6%	0.5154	`fda_pe`
MeSH actions	NLM's indexed pharmacological actions	many actions name a mechanism	1,028	865	45.0%	0.5945	`mesh_pa` (SIMILARITY-HARNESS §10, §12)
label effects	five nearest label neighbours	**nothing**	409	346	14.5%	0.2620	`label_effects`; `onsides_key.json`

key	judges	shares with the score	questions	graded	top-1	MRR	where
pairs	243 sourced pairs	the mechanisms labels and reviews state	243 pairs	—	—	—	`pairs.json`; **held out**

The grades are not comparable across rows as absolute numbers. A key with broad classes or many classmates per question is easier to hit, which is why `chance.mjs` sets each grade beside a shuffle of the same lists. Against the shuffle, every key puts the ranking's top-1 at about five to nine times chance.

How the keys are used together. SIMILARITY-LITERATURE §4.1 asks that a scoring constant be changed only when judges of different provenance agree, with one key held out. In the sweep across five judges (SIMILARITY-HARNESS §15), every judge rates switching off the opposite-direction demotion (D15) as the worst single setting. Lowering the weight of replicated-only claims helps only under the judges that read curation (ATC, MeSH) and hurts under the label key. That is the pattern the label key was built to catch.

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7. Reading a grade

- **Question.** A molecule the key can ask about.
- **Scoreable / unanswerable.** A question is unanswerable when none of its classmates appears anywhere in its similarity list, because they share no target in ChEMBL. Unanswerable questions are counted apart and left out of every rate. Otherwise, a change that made more claims eligible would raise the grade by adding answerable questions, not by ranking better.
- **Top-1.** The share of graded questions whose first answer is a classmate.
- **MRR** (mean reciprocal rank). Each question scores 1 ÷ the rank of its first classmate, and the scores are averaged. Rank 1 scores 1.0, rank 2 scores 0.5 and rank 10 scores 0.1. It notices a move from rank 50 to rank 8, which top-1 does not.
- **Filtered.** The same list, walked past only the molecules the key can judge. Seltorexant above suvorexant's classmates is neither right nor wrong under ATC, which has no code for it, so the filtered reading steps over it. This is the "filtered" setting of link-prediction evaluation. It changes the judge, not the ranking, and it is not an upper bound.
- **nDCG@10** (normalised discounted cumulative gain over the top ten). It credits every relevant molecule in the first ten, not just the first, discounts lower ranks, and allows half credit (a sibling class, a MoA match, a label neighbour sixth to tenth).
- **Against a shuffle.** The grade the same list would get in random order, computed in closed form (`chance.mjs`). A grade means something only as its distance from that figure.
- **Held out.** A key no sweep may tune to. Only the pairwise key is held out.
- **Fixed / frozen, versioned.** Each key's rules sit in a JSON file with a version and a date. Changing it is a change of instrument, recorded in SIMILARITY-HARNESS.md and never made in the same change as a constant it judges.

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8. Commands and files

```
        node analysis/similarity_harness/evaluate.mjs                      # ATC, the headline
        node analysis/similarity_harness/evaluate.mjs --key atc-repaired
        node analysis/similarity_harness/evaluate.mjs --key epc            # also --key pe
        node analysis/similarity_harness/evaluate.mjs --key onsides        # also onsides-all
        node analysis/answer_keys/pairs.mjs                                # the pairwise key
        node analysis/answer_keys/neighbours.mjs SERTRALINE                # what the label key
says
        node analysis/answer_keys/agreement.mjs                            # how far the keys
agree
        node analysis/answer_keys/chance.mjs                               # each grade beside a
shuffle

        python3 scripts/resolve_cohort_fda_classes.py     # FDA classes via RxNav (~50 min cold,
cached)
        python3 scripts/resolve_cohort_label_effects.py   # OnSIDES label effects; after the FDA
classes
```

file	what it holds
`analysis/similarity_harness/harness.mjs`	every key's construction and the grading walk
`analysis/similarity_harness/atc_repaired.json`	the repaired ATC key's exclusions, each with its reason
`analysis/similarity_harness/fda_keys.json`	which MoA classes earn no credit, and why
`analysis/similarity_harness/onsides_key.json`	the label key's rules, each with its reason
`analysis/answer_keys/pairs.json`	the 243 pairs, their sources, rules and provenance
`scripts/resolve_cohort_fda_classes.py`	the RxNav route, in its docstring
`scripts/resolve_cohort_label_effects.py`	OnSIDES and ADReCS, in its docstring

Further reading: SIMILARITY-HARNESS.md for the harness and every measurement; SIMILARITY-LITERATURE.md for why these keys were chosen, the literature behind them and what ATC gets wrong; DATA-QUALITY-LOG.md, 2026-10-13 and 2026-10-14, for what went wrong building the FDA and OnSIDES backfills.